

NumPy Probability Distributions — Line by Line Examples

This PDF explains Uniform, Normal, Binomial, Poisson, Exponential, and Standard Normal distributions using simple NumPy examples. Outputs involving random numbers may differ unless a seed is used.

1. Uniform Distribution — `np.random.uniform()`

Simple explanation: Uniform distribution gives random values where every value in the selected range has equal chance.

Code:

```
import numpy as np

np.random.seed(1)

result = np.random.uniform(10, 20, 5)

print(result)
```

Output:

```
Output:
[14.17022005 17.20324493 10.00114375 13.02332573 11.46755891]
```

Important: 10 = lower limit, 20 = upper limit, 5 = number of random values.

2. Normal Distribution — `np.random.normal()`

Simple explanation: Normal distribution is centered around a mean. The standard deviation controls how spread out the values are.

Code:

```
import numpy as np

np.random.seed(1)

result = np.random.normal(
    loc=50,
    scale=10,
    size=5
)

print(result)
```

Output:

```
Output:
[66.24345364 43.88243586 44.71828248 39.27031378 58.65407629]
```

Important: loc=50 means mean = 50. scale=10 means standard deviation = 10. size=5 creates 5 values.

3. Binomial Distribution — `np.random.binomial()`

Simple explanation: Binomial distribution counts the number of successes in a fixed number of independent trials.

Code:

```
import numpy as np

np.random.seed(1)
```

```
result = np.random.binomial(
    n=10,
    p=0.5,
    size=5
)

print(result)
```

Output:

```
Output:
[5 7 0 4 4]
```

Important: $n=10$ means 10 trials. $p=0.5$ means 50% probability of success. Each output is the number of successes.

4. Poisson Distribution — `np.random.poisson()`

Simple explanation: Poisson distribution models how many times an event happens in a fixed interval when an average rate is known.

Code:

```
import numpy as np

np.random.seed(1)

result = np.random.poisson(
    lam=4,
    size=5
)

print(result)
```

Output:

```
Output:
[2 2 3 3 4]
```

Important: $\text{lam}=4$ means the average number of events is 4 per interval. $\text{size}=5$ creates 5 simulated intervals.

5. Exponential Distribution — `np.random.exponential()`

Simple explanation: Exponential distribution is commonly used for waiting time between events.

Code:

```
import numpy as np

np.random.seed(1)

result = np.random.exponential(
    scale=2,
    size=5
)

print(result)
```

Output:

```
Output:
[1.07921167 2.54825051 0.00022875 0.72014422 0.31741928]
```

Important: $\text{scale}=2$ is the average waiting time. $\text{size}=5$ creates 5 waiting-time samples.

6. Standard Normal Distribution — np.random.randn()

Simple explanation: Standard normal distribution has mean 0 and standard deviation 1.

Code:

```
import numpy as np

np.random.seed(1)

result = np.random.randn(5)

print(result)
```

Output:

```
Output:
[ 1.62434536 -0.61175641 -0.52817175 -1.07296862  0.86540763]
```

Important: np.random.randn(5) generates 5 values from a standard normal distribution.

7. Mean

Simple explanation: Mean is the average of values. Add all values and divide by the number of values.

Code:

```
import numpy as np

data = np.array([10, 20, 30, 40, 50])

result = np.mean(data)

print(result)
```

Output:

```
Output:
30.0
```

Important: Calculation: $(10 + 20 + 30 + 40 + 50) / 5 = 30$.

8. Standard Deviation

Simple explanation: Standard deviation tells us how far values are spread from the mean.

Code:

```
import numpy as np

data = np.array([10, 20, 30, 40, 50])

result = np.std(data)

print(result)
```

Output:

```
Output:
14.142135623730951
```

Important: A small standard deviation means values are close to the mean. A large standard deviation means values are more spread out.

9. Probability

Simple explanation: Probability tells us how likely an event is. It ranges from 0 to 1, or 0% to 100%.

Code:

```
# Example: probability of getting a 6 on a fair die

probability = 1 / 6

print(probability)
print(probability * 100)
```

Output:

```
Output:
0.16666666666666666
16.666666666666664
```

Important: There is 1 favorable result out of 6 possible results, so probability = 1/6.

10. Random Sampling

Simple explanation: Random sampling means selecting observations randomly from a population.

Code:

```
import numpy as np

np.random.seed(1)

data = np.array([10, 20, 30, 40, 50])

sample = np.random.choice(data, size=3, replace=False)

print(sample)
```

Output:

```
Output:
[30 20 50]
```

Important: np.random.choice() selects random values. replace=False means the same value cannot be selected twice.

11. Simulation

Simple explanation: Simulation means using random numbers to imitate a real-world process.

Code:

```
import numpy as np

np.random.seed(1)

coin_flips = np.random.choice(
    ["Head", "Tail"],
    size=10
)

print(coin_flips)
```

Output:

```
Output:
['Tail' 'Tail' 'Head' 'Head' 'Tail' 'Tail' 'Tail' 'Tail' 'Tail' 'Head']
```

Important: Here, 10 random coin-flip results simulate 10 coin tosses.

Quick Reference

np.random.uniform() — Random values in a range; equal chance across the range.

np.random.normal() — Random values based on mean and standard deviation.

np.random.binomial() — Number of successes in fixed trials.

np.random.poisson() — Number of events for a given average rate.

np.random.exponential() — Waiting time between events.

np.random.randn() — Standard normal values: mean 0, standard deviation 1.

Mean — Average value.

Standard deviation — Amount of spread around the mean.

Probability — Chance that an event happens.

Random sampling — Selecting observations randomly.

Simulation — Using random data to imitate a real process.